Assignment 3

10 pts



جامعة قطر
QATAR UNIVERSITY

**CSE Department**
**Computer Vision CMAI 480**

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**Assignment Instructions:**

- The assignment must be turned electronically using Blackboard.
- This is an individual assignment -- each student should submit their own solution.
- In every uploaded file, include your name, date, and the assignment name.
- Use MATLAB for all tasks. Your script must run from beginning to end without manual repair.
- Use the supplied image pair, image1.png and image2.png, unless instructed otherwise.
- No makeups for the assignment are allowed.
- Please start early and ask me for help if you get stuck.

# Assignment 3: SIFT Feature Detection, Matching, and Image Registration using RANSAC

## Overview

- Detect scale-invariant feature points and extract local descriptors from two overlapping images.
- Match feature descriptors and reject ambiguous correspondences using a ratio-based criterion.
- Estimate a projective transformation (homography) robustly in the presence of outliers using a RANSAC-family estimator.
- Visualize putative matches and geometrically consistent inlier matches.
- Register one image with the other and combine both images into a single panorama-style result (or simply into a Combined registered image).
- Evaluate registration quality using match counts, inlier ratio, visual alignment, and discussion.

## Requirements

- MATLAB with Image Processing Toolbox.
- Computer Vision Toolbox, including SIFT feature detection, feature extraction, matching, and geometric-transform estimation functions.
- Two overlapping images of the same scene with sufficient texture and shared content.

## Tasks

### 1. Load and Validate the Images

Load image1.png and image2.png using imread. Confirm that both files exist, display them side-by-side, and report their sizes and data types.

**You can use the following MATLAB starting code segment:**

```
I1 = imread('image1.png');
I2 = imread('image2.png');
figure; montage({I1,I2}); title('Input Images');
```

### 2. Convert the Images to Grayscale

Convert both images to grayscale. If an input is already grayscale, keep it unchanged. SIFT feature detection and descriptor extraction will use these grayscale images.

**You can use the following MATLAB starting code segment:**

```
G1 = im2gray(I1);
G2 = im2gray(I2);
```

### 3. Detect and Visualize SIFT Features

Detect SIFT features in both images. Report the number of detected points and visualize the strongest points, including their scale and orientation.

**You can use the following MATLAB starting code segment:**

```
P1 = detectSIFTFeatures(G1);
P2 = detectSIFTFeatures(G2);
figure; imshow(I1); hold on; plot(P1.selectStrongest(150));
```

### 4. Extract Descriptors and Match Features

Extract descriptors at the valid SIFT points. Match the descriptors and display the putative correspondences before geometric verification.

**You can use the following MATLAB starting code segment:**

```
[F1,V1] = extractFeatures(G1,P1);
[F2,V2] = extractFeatures(G2,P2);

% You can experiment with the ratio below so that you find the best matches
indexPairs = matchFeatures(F1,F2,MaxRatio=0.75,Unique=true);
M1 = V1(indexPairs(:,1)); M2 = V2(indexPairs(:,2));
```

### 5. Filter and Evaluate Putative Matches

Use the descriptor ratio criterion in matchFeatures to remove ambiguous matches. Experiment reasonably with MaxRatio, explain the chosen value, and report the number of retained matches. At least four non-collinear correspondences are required for a projective transform.

**You can use the following MATLAB starting code segment:**

```
fprintf('Putative matches: %d\n',size(indexPairs,1));
figure; showMatchedFeatures(I1,I2,M1,M2,'montage');
```

### 6. Estimate a Projective Transformation Using RANSAC

Estimate a homography that maps points in Image 1 to Image 2. Use estgeotform2d with the projective model. The estimator uses MSAC (M-estimator SAmple Consensus), a robust variant of RANSAC (RANdom SAmple Consensus), to reject outliers. Print the transformation matrix, identify inliers, compute the inlier ratio, and display inlier matches only.

**You can use the following MATLAB code segment:**

```
% You may need to experiment with parameters set below

[tform,inlierIdx] = estgeotform2d(M1,M2,'projective', ...
    MaxDistance=4,Confidence=99.9,MaxNumTrials=5000);

inlier1 = M1(inlierIdx);

inlier2 = M2(inlierIdx);

fprintf('Inliers: %d of %d
(%.2f%%)\n',nnz(inlierIdx),numel(inlierIdx),100*mean(inlierIdx));

disp('Projective transformation matrix:');

disp(tform.A);

figure('Name','Inlier Matches');

showMatchedFeatures(I1,I2,inlier1,inlier2,'montage');

title('Geometrically Consistent Inlier Matches');
```

## 7. Register the Images

Warp Image 1 into the coordinate system of Image 2 using the estimated projective transformation. Compute an output spatial reference large enough to contain both images and display the registered result.

**You can use the following MATLAB code segment:**

```
[h1,w1,~]=size(I1);
[h2,w2,~]=size(I2);
[xlim1,ylim1]=outputLimits(tform,[1 w1],[1 h1]);
xMin=floor(min([1 xlim1]));
xMax=ceil(max([w2 xlim1]));
yMin=floor(min([1 ylim1]));
yMax=ceil(max([h2 ylim1]));
width=xMax-xMin+1;
height=yMax-yMin+1;
CombinedImg=imref2d([height width],[xMin xMax],[yMin yMax]);
identity=affinetform2d(eye(3));
W1=imwarp(I1,tform,OutputView=CombinedImg);
W2=imwarp(I2,identity,OutputView=CombinedImg);
mask1=imwarp(true(h1,w1),tform,OutputView=CombinedImg);
mask1_3 = repmat(mask1,[1 1 3]); % mask to all three color channels
figure('Name','Registered Image 1');
imshow(W1); title('Image 1 Registered to Image 2 Coordinates');
```

## 8. Combine and Display the Registered Images

Combine the registered images into a single output image by placing both registered images in the same coordinate system. Display the resulting combined image and evaluate the registration quality visually. No image blending is required. Display and save the final image as Combined.jpg.

**You can use the following MATLAB code segment:**

```
% Create empty output image
combined = W2;
% Fill pixels from the transformed image
combined(mask1_3) = W1(mask1_3);
figure('Name','Combined Registered Image');
imshow(combined);
title('Combined Registered Image');
imwrite(combined,'Combined.jpg');
```

## 9. Analyze the Results

Report the numbers of detected features, putative matches, and inlier matches. Calculate the inlier ratio. Discuss alignment in the overlap, visible seams or ghosting, likely causes of incorrect matches, and one reasonable improvement.

**You can use the following MATLAB code segment:**

```
inlierRatio = nnz(inlierIdx)/numel(inlierIdx);
fprintf('\nSummary: features=(%d,%d), putative=%d, inliers=%d, ratio=%.2f%%\n', ...
    P1.Count,P2.Count,size(indexPairs,1),nnz(inlierIdx),100*inlierRatio);
```

## 10. Testing the Registration Pipeline on a Challenging Image Pair

The provided images **image3.png** and **image4.png** form a challenging image pair for the featue-based registration method developed in this assignment. Consequently, Modify your solution so that it now processes:

- image3.png
- image4.png

instead of the original image pair. Run the complete registration pipeline, and **report** the following numerical results as comments in the solution script:

1. Number of SIFT features detected in each image
2. Number of putative matches
3. Number of RANSAC/MSAC inliers
4. Inlier ratio
5. Whether a valid and visually meaningful projective transformation was obtained
6. If the registration fails or produces an inaccurate result, explain the likely reasons. Your explanation should consider several possible factors, including:
    - Limitations of SIFT feature detection and description
    - Insufficient distinctive texture
    - Repeated or visually similar structures within the scene
    - Insufficient overlap between the images
    - Significant viewpoint or perspective changes
    - Parallax caused by objects at different depths
    - Differences in camera position, camera orientation, or focal length
    - Image blur, noise, compression, or low resolution
    - Changes in illumination, shadows, reflections, or moving objects
    - Limitations of representing the entire scene using a single projective homography
    - Do not simply state that “SIFT failed.” Identify the stage or stages of the pipeline at which the failure occurred and support your explanation using the displayed results and numerical measurements.

## 11. Conceptual Questions

1. Why are SIFT features suitable for matching images with changes in scale and rotation?
2. What is the purpose of the descriptor ratio criterion, and what is the trade-off when its threshold is made stricter?
3. Why is ordinary least-squares estimation unreliable when feature correspondences contain outliers?
4. What is the difference between a putative match and a RANSAC/MSAC inlier?

5. Under what scene and camera assumptions is a single projective homography appropriate?

## Deliverables

- A completed MATLAB script named Assignment3.m, divided into executable sections beginning with %%.
- Keypoint visualizations for both images.
- A visualization of putative feature matches.
- A visualization of RANSAC/MSAC inlier matches.
- The registered image and the final combined image named Combined.jpg.
- A PDF generated from the completed MATLAB script, showing code, figures, numerical results, and answers.
- One ZIP archive containing all required files.